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Recent Legislative Changes Have Contributed to Growth in Ohio's Energy Production

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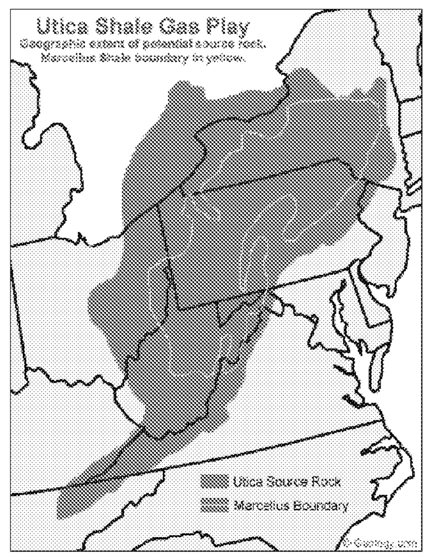
Following a change in state legislation in 2023, the Ohio Oil and Gas Land Management Commission (OGLMC) began awarding bids to energy developers for exploration on state-controlled land. Advances in horizontal drilling and hydraulic fracturing (fracking) had already propelled the state to become the nation's sixth-largest natural gas producer and twelfth-largest crude oil producer. While growth in energy production has not increased employment and gross state product as was forecast, tax revenues have risen in mining-related areas and counties heavily affiliated with the energy sector. These effects and other demographic and economic trends in mining-dependent counties may affect bank deposit and loan growth patterns.

Map 1



Although Ohio has allowed fracking for years, the state took action in 2011 and again in 2023 to allow more drilling on state-owned lands. These lands include parts of Eastern Ohio that are in the Appalachian Basin, home to the Marcellus and Utica Shale formations, which have the potential to supply large amounts of natural gas and crude oil (Map 1). In 2011, Ohio passed House Bill 133 to create the OGLMC, responsible for allowing drilling on state land.[1] In January 2023, the state government passed House Bill 507, which changed the language regarding leases granted by the OGLMC to speed up the process for energy developers to lease land.[2] In February 2024, the OGLMC started to award bids related to drilling in various state parks.

With increased fracking, Ohio's oil and natural gas production has risen substantially. Crude oil and natural gas production in Ohio, a mid-ranked energy producer in the 2000s, rose sharply as new drilling techniques that combined horizontal drilling with fracking made shale deposits more accessible, similar to trends in other Utica and Marcellus shale states.

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[1] House Bill Number 133, 129th General Assembly.

[2] House Bill Number 507, 134th General Assembly. The text of the bill states, "a state agency shall lease, in good faith, [land] that is owned or controlled by the state agency for the exploration for and development and production of oil and natural gas." The previous law did not mandate the leasing.

In the 2000s, the state averaged slightly less than 1.4 million barrels of crude oil quarterly, 0.28 percent of the nation's production, with production largely flat over the decade. In the decade following 2010, the state's energy production increased dramatically, with the average production value nearly three times as high as the previous decade. Crude oil production in the state hit an all-time high of 8.5 million barrels in fourth quarter 2023, rising to nearly 0.69 percent of the energy production in the United States (Chart 1). Ohio rose to the 12th-highest producer of oil in the nation from 16th a decade earlier, and the highest-producing state in the Appalachian Basin in 2023, which includes nine states spanning from Alabama to Maine.

Ohio's natural gas production, which averaged more than 23 million cubic feet per quarter before 2010, similarly increased to average production of 255 million cubic feet per quarter in the 2010s. As exploration activity grew, production increased dramatically before settling closer to 500 million cubic feet quarterly.[3] The state produced a little more than 559 million cubic feet in fourth quarter 2023, making Ohio the sixth-highest producer of natural gas in the country, up from 21st in 2010 (Chart 2).

Chart 1

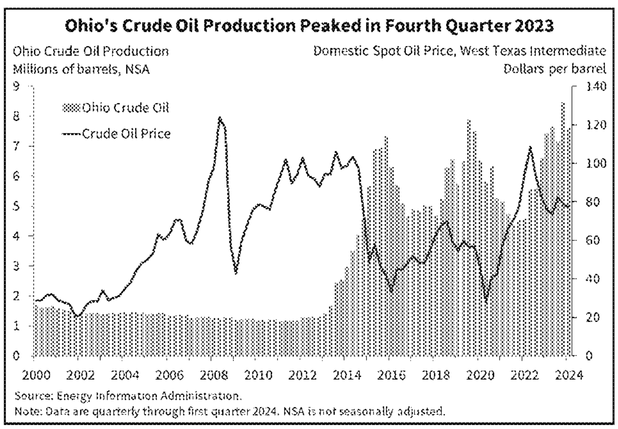
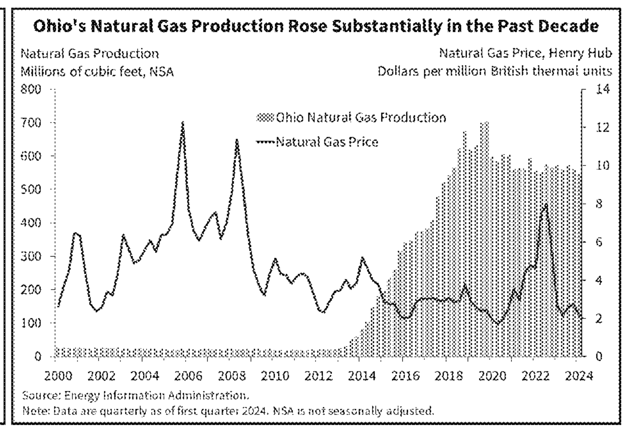


Chart 2



Higher crude oil prices and declining natural gas prices likely affected energy exploration and production. Higher crude oil prices in the 2010s and 2020s may have contributed to increased oil exploration and production. From 2000 to 2010, a barrel of crude oil was trading at a median of roughly \$51. From 2011 to 2024, the median price per barrel rose nearly 40 percent to slightly less than \$71. Conversely, prices may not have had a similar effect on natural gas production in the state. The median price of natural gas fell from about \$6 per million British thermal units (MMBTU) in the 2000s to less than \$3 per MMBTU from 2011 to 2024.

[3] Ohio's 500 million cubic feet produced is roughly half of the quarterly production of Louisiana, the third-highest producer of natural gas in the country. The first two are Texas and Pennsylvania.

Experience suggests increased exploration on state-owned land is unlikely to affect the economy but may increase tax revenue. When the 2011 legislation was passed, an industry forecast predicted that energy exploration in Ohio would lead to large increases in gross state product (GSP), employment, and tax revenue.[4] The report forecast more than 4,000 new wells, an increase of more than 11 percent, between 2010 and 2015; GSP to increase more than 600 percent, or $12 billion; and mining support activity employment to add more than 117,000 jobs, which would put the sector at just more than 2 percent of the state's total employment.[5] State tax revenues were projected to increase $479 million, or 2 percent, led by increases in income taxes and smaller gains in property, commercial activity, and severance taxes.

The actual effects on Ohio's GSP and employment were considerably less than the 2011 industry forecast. Total contributions from the mining sector to Ohio's GSP rose slightly more than $3 billion, or 158 percent, and remained less than 1 percent of total state economic output. Employment in the support activities for mining subcategory increased by less than 3,000 jobs between 2010 and 2015. In the following years, employment in the mining sector generally declined and struggled to return to its peak of slightly less than 16,000 jobs in 2014 (Chart 3). While increased fracking and energy exploration did not result in significant GSP and employment growth, Ohio tax revenue related to mining increased considerably. Commercial activity, property, sales and use, and severance tax revenues grew, in some cases considerably (Chart 4).

Chart 3

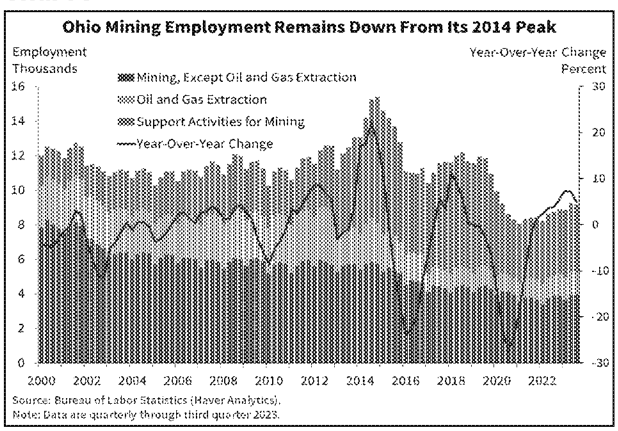
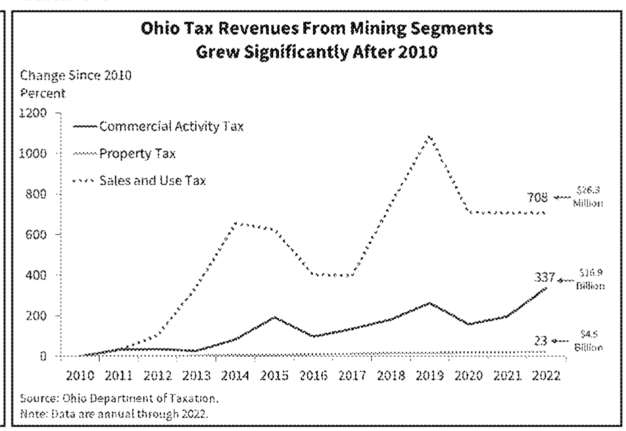


Chart 4



Commercial activity tax revenue from the mining sector grew from $3.8 billion in 2010 to nearly $17 billion in 2022, a 337 percent increase. Property tax revenue from commercial, industrial, and mineral properties in the state grew more than 23 percent in the same period, reaching more than $4.4 billion in 2022. Growth was much higher in counties that are more dependent on the mining

[4] Jack Kleinhenz and Russ Smith, "Ohio's Natural Gas and Crude Oil Exploration and Production Industry and the Emerging Utica Gas Formation, Economic Impact Study," Kleinhenz & Associates, September 2011.
[5] In 2010, mining support activity employment was only 0.05 percent of total employment.

sector, as property tax revenues there grew more than 200 percent (Chart 5).[6] Sales and use tax revenue related to mining increased from $3.3 million in 2010 to $26.3 million in 2022, a 708 percent increase. While severance tax revenue increased only $17 million from 2010 to 2015—less than the $32 million forecast for the period—the category grew more than 145 percent and continued to grow throughout the rest of the 2010s before leveling off (Chart 6).

Chart 5

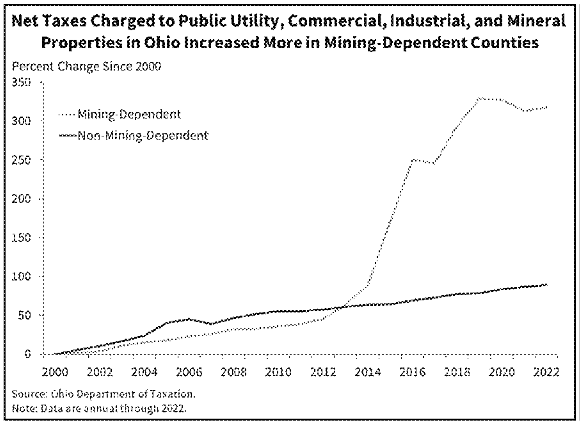
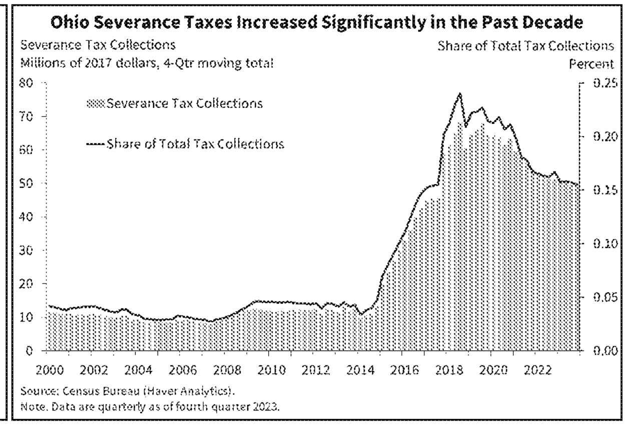


Chart 6



Some critics of Ohio's expansion of energy exploration on public lands have argued that its low severance tax rate fails to compensate the public for potential harm from drilling.[7] Previous attempts to increase the state's severance tax rate were defeated in the state legislature. However, leases awarded by the OGLMC for exploration on public lands require at least a 12.5 percent royalty payment to the state based on the value of production, five times the current severance tax rate of 2.5 cents per million cubic feet of production.

Ohio's mining-dependent counties underperformed their counterparts despite tax growth. Other analyses have shown that mining-dependent counties had greater economic performance than counties not as heavily centered on the industry. This has not appeared to be the case in Ohio over the past two decades, as some measures of economic performance are weaker in mining-dependent counties (Table 1). Ohio counties that are more dependent on mining have had an unemployment rate nearly 2 percentage points higher than other counties. Population in these counties grew at a lower rate from 2000 to 2005 and fell even further after 2005, as the rest of Ohio continued to grow. Personal income has grown in both types of counties since 2000, though that growth has been slightly weaker in mining-dependent counties.

[6] This article identifies mining-dependent counties as those where the mining industry produces 15 percent or more of the county gross product. For a similar analysis, see Jeff Ayres et al., "Mining Activity Boosts Economic Growth in Energy-Rich Areas and Benefits Community Banks," *Perspectives*, July 2013.
[7] Marty Schladen, "Oil and Gas Drilling in State Parks Set to Expand, While Some Say Taxes on It Are Too Low" *Ohio Capital Journal*, April 21, 2023.

Table 1

Demographic and Economic Indicators Were Slightly Weaker in Ohio Mining Counties Since 2000

	Mining-Dependent	Non-Mining-Dependent
Total Employment Growth	-13.2%	-2.3%
Mining Employment Growth	30.5%	-4.9%
Non-Mining Employment Growth	-42.3%	-1.3%
New Business Formation Growth	-3.8%	13.7%
Population Growth	-9.0%	4.0%
Personal Income Growth	94.4%	109.9%

Sources: Bureau of Labor Statistics, Bureau of Economic Analysis, Census Bureau (Haver Analytics).
Note: Population and employment growth rates are calculated over the period 2000 to 2023. Personal income data are calculated over the period 2000 to 2022, the most recent data available.

Economic conditions in mining counties could have been weaker over the past decade if horizontal drilling and fracking had not occurred. A 2023 report from the Ohio River Valley Institute compared economic performance of rural counties in Pennsylvania, another Utica and Marcellus Shale state, based on their participation in the natural gas boom of the 2010s.[8] The report found both types of counties had similar trajectories in employment, population, and per capita GDP before and after the sector's upturn in 2010, while counties that did not participate in mining saw a slightly higher decline in total employment. Referring to residents of mining-dependent counties, Ohio State Representative Jack Cera said, "We had nothing going on down here; fracking is a lifesaver for a lot of people. There was an influx into the economy that would have never happened" without fracking.[9]

Demographic and economic trends may have affected deposit and loan growth in mining-dependent-county banks. Banks in both types of counties have shown similar trends in deposit and loan growth, although measures in mining-dependent counties have been more volatile. Deposits in a typical bank in a non-mining-dependent county have grown consistently since 2000, while deposit growth in a typical bank in a mining-dependent county has been more volatile, with some declines in the past two decades (Chart 7). Similarly, a typical bank in a non-mining-dependent county had stable loan growth for most of the same period, while banks in mining-dependent counties had similar volatility in loan growth (Chart 8).

[8] Sean O'Leary, "Frackalachia Update: Peak Natural Gas and the Economic Implications for Appalachia," Ohio River Valley Institute, August 22, 2023.
[9] Eric Heisig, "Ohio's Fracking Industry—Boon as Promised or Something Less?" Cleveland.com, March 19, 2022.

Chart 7

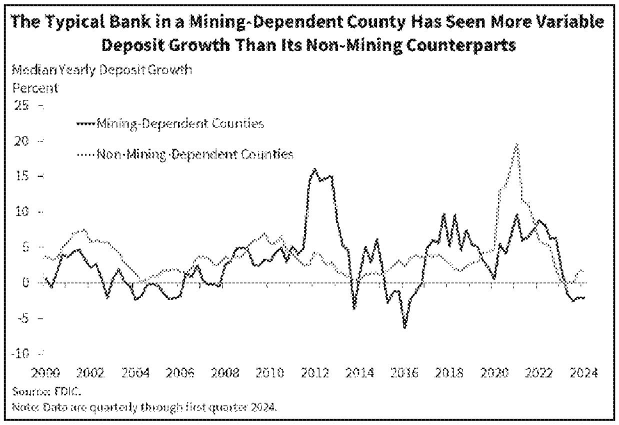
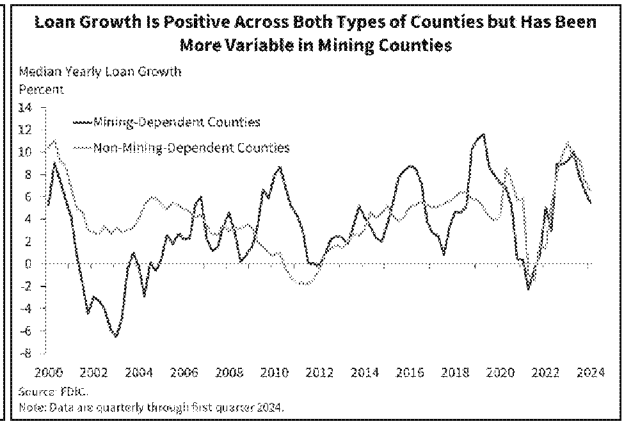


Chart 8



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